

Part C

Course title: Bayes Methods
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1. In a study of the association of DNA mutations with hair color, DNA sequences were observed for n individuals and their hair color recorded. For $i = 1, 2, \dots, n$ let $y_i = 1$ if individual i has red hair and $y_i = 0$ if not, let $x_i = (x_{i,1}, \dots, x_{i,L})$ with $x_{i,j} = 0$ if there is no mutation at site $j = 1, 2, \dots, L$ and $x_{i,j} = 1$ if there is a mutation. Let $X = [x_{i,j}]_{i=1:n, j=1:L}$. The number of sites $L \gg 1$ is large.

Let α be an intercept parameter and $\beta = (\beta_1, \dots, \beta_L)$ a parameter vector of mutation effects. Let $z = (z_1, \dots, z_L)$ be a binary model indicator vector $z \in \Omega_z$ with $\Omega_z = \{0, 1\}^L$. Let $\beta_z = (z_1\beta_1, \dots, z_L\beta_L)^T$. Consider a logistic regression of y on X given z , $y_i \sim \text{Bernoulli}(p_i)$ with $p_i(\alpha, \beta, z) = \exp(\alpha + x_i\beta_z) / (1 + \exp(\alpha + x_i\beta_z))$.

(a) [10 marks]

- (i) Write down the posterior $\pi(\alpha, \beta, z|y)$ in as much detail as you can, but without eliciting a prior for $\pi(\alpha, \beta, z)$.
- (ii) It is of interest to see if any of the mutations are associated with hair color. If some of the mutations are associated with hair color then it is expected that the number will be small, perhaps four or five. Write down a prior $\pi(z)$ for z informed by this information.
- (iii) You have access to a second large and completely independent data set of sequence data $\tilde{X} = [\tilde{x}]_{i=1:N, j=1:L}$ for individuals $i = 1, 2, \dots, N$ sampled at random from a large population the hair color is not recorded, but the proportion μ of red heads in the sampled population is known. Briefly, how does the data $\mu, \tilde{X}$ inform a prior for α, β given $\pi(z)$? How would you use simulation to validate the joint prior for α, β, z ?

(b) [9 marks] Suppose $\pi(\alpha, \beta|z) = N(\alpha; 0, 1) \prod_{i=1}^n N(\beta_i; 0, 1)$ (ie, independent of z). Notice that we interpret $\beta_i z_i$ as the effect due to a mutation at site i , not β_i by itself.

- (i) Give a MCMC algorithm targeting $\pi(\beta, z|y)$, simplifying as far as possible.
- (ii) How would you test for significant effects? Is there a hazard for multiple testing in your approach? Justify your answer.

(c) [6 marks]

- (i) The analysis is restricted to models $z \in A$ with $A \subseteq \Omega_z$. How would you modify your MCMC to simulate $\pi(z|z \in A, y)$? State any assumptions concerning A .
- (ii) Let B_A be the Bayes factor testing $z \in A$ against $z \in \Omega_z$. Show that

$$B_A = \frac{\pi(A|y)}{\pi(A)},$$

where $\pi(A|y) = \sum_{z \in A} \pi(z|y)$ and $\pi(A) = \sum_{z \in A} \pi(z)$ and state briefly how you would estimate this.

2. For $i = 1, 2, \dots, n$ let $Y_i \sim f(y_i|x_i)$ when $X = x$ with $x = (x_1, \dots, x_n)$. Suppose $X \sim \pi(x|\theta)$ and $\theta \sim \pi(\theta)$ with θ a real scalar parameter.

(a) [15 marks] Suppose we observe $X = x$ and wish to estimate θ .

- (i) Explain what it means for $\pi(\theta|x)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $X = x$.
- (ii) Let $S(x) = (S_1(x), \dots, S_p(x))$ be a p -component vector of statistics computed from the data. For $S, S' \in \mathbb{R}^p$ let $d(S, S')$ be a distance function and let $\rho > 0$ be a simulation parameter. Write down an ABC algorithm simulating

$$\theta \sim \pi(\theta|d(S(X), S(x)) < \rho)$$

and prove that your algorithm targets $\pi(\theta|d(S(X), S(x)) < \rho)$.

- (iii) An ABC algorithm yields samples $\theta^{(t)}, t = 1, 2, \dots, T$ with associated statistics $S^{(t)} = (S_1^{(t)}, \dots, S_p^{(t)})$. Explain how to make a regression adjustment to the sampled θ -values. What is the purpose of this adjustment?
- (b) [10 marks] Now suppose $Y = y$ with $y = (y_1, \dots, y_n)$ is observed and X and θ are unknown. The Ising model is a random process for binary $m \times m$ images. For $i = 1, 2, \dots, n$ with $n = m^2$, $x_i \in \{0, 1\}$. Let N_i be the set of neighbors of pixel i on the square lattice. Let $\#x$ denote the number of disagreeing neighbors, that is

$$\#x = \frac{1}{2} \sum_{i=1}^{m^2} \sum_{j \in N_i} \mathbb{I}_{x_i \neq x_j}.$$

Under the Ising model, $\pi(x|\theta) = \exp(-\theta\#x)/Z(\theta)$ where $Z(\theta)$ is a normalising constant. Let $y_i \sim N(x_i, \sigma^2)$ for $i = 1, 2, \dots, n$ with $n = m^2$. A prior $\theta \sim \text{Exp}(2)$ is elicited for θ .

- (i) Write down the posterior $\pi(\theta, x|y)$ in terms of the model elements and explain why it is doubly intractable when $m \gg 1$.
- (ii) Consider the statistic

$$S(y) = \frac{1}{2} \sum_{i=1}^{m^2} \sum_{j \in N_i} (y_i - y_j)^2.$$

Briefly motivate this choice of the ABC statistic.

- (iii) An MCMC algorithm for $X \sim \pi(x|\theta)$ is available. Give an ABC algorithm targeting $\pi(\theta|d(S(Y), S(y)) < \rho)$ using the MCMC algorithm to simulate $X \sim \pi(x|\theta)$.
- (iv) Explain briefly why your algorithm targeting $\pi(\theta|d(S(Y), S(y)) < \rho)$ will not in general be exact.

3. For $i = 1, 2, \dots, n$ let $y_i \sim f(y_i|\theta_i)$ where $\theta_i \in \Omega$. A random distribution $G \sim DP(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H on Ω . Given G , the θ_i , $i = 1, 2, \dots, n$ are independent and identically distributed according to G . Let $\theta = (\theta_1, \dots, \theta_n)$ and let $f(y|\theta) = \prod_{i=1}^n f(y_i|\theta_i)$.

(a) [2 marks] Give a definition of a Dirichlet Process.

(b) [7 marks] Show that for sets $A \subseteq \Omega$,

$$\Pr(\theta_{n+1} \in A|\theta) = \frac{\alpha H(A) + \sum_{i=1}^n \mathbb{I}_{\theta_i \in A}}{\alpha + n}.$$

You may assert without proof that if $G \sim DP(\alpha, H)$ and $(\theta_n|G) \sim G$ then $\theta_n \sim H$.

- (c) [7 marks] Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ list the distinct values in θ , with $K \in \{1, 2, \dots, n\}$. Let $S = (S_1, \dots, S_K)$ be the partition of $\{1, 2, \dots, n\}$ which satisfies $i \in S_k$ if $\theta_i = \theta_k^*$ and let $|S_k| = \text{card}(S_k)$. Let $P(S)$ be the probability θ has partition S . Show that

$$P(S) = \frac{\alpha^K \prod_{k=1}^K (|S_k| - 1)}{\prod_{i=1}^n (\alpha + i - 1)}.$$

- (d) [7 marks] Suppose that for $k = 1, 2, \dots$, $H(d\theta_k^*) = h(\theta_k^*)d\theta_k^*$. Let $h(\theta^*) = \prod_{k=1}^K h(\theta_k^*)$ and let $S^{-i} = (S_1^{-i}, \dots, S_K^{-i})$ denote the partition S with i removed. Explain why

$$\pi(d\theta|y) \propto f(y|\theta^*, S)h(d\theta^*)P(S)$$

and show that

$$\Pr(i \in S_k|S^{-i}, \theta^*) = \begin{cases} |S_k^{-i}|f(y_i|\theta_k^*) & \text{if } k \in \{1, 2, \dots, K\} \\ \alpha f(y_i|\theta_k^*) & \text{if } k = K + 1. \end{cases}$$

- (e) [2 marks] Suppose $h(\theta)$ is some general (ie, non-conjugate) prior on Ω . Briefly outline an MCMC algorithm targeting $\pi(\theta|y)$.